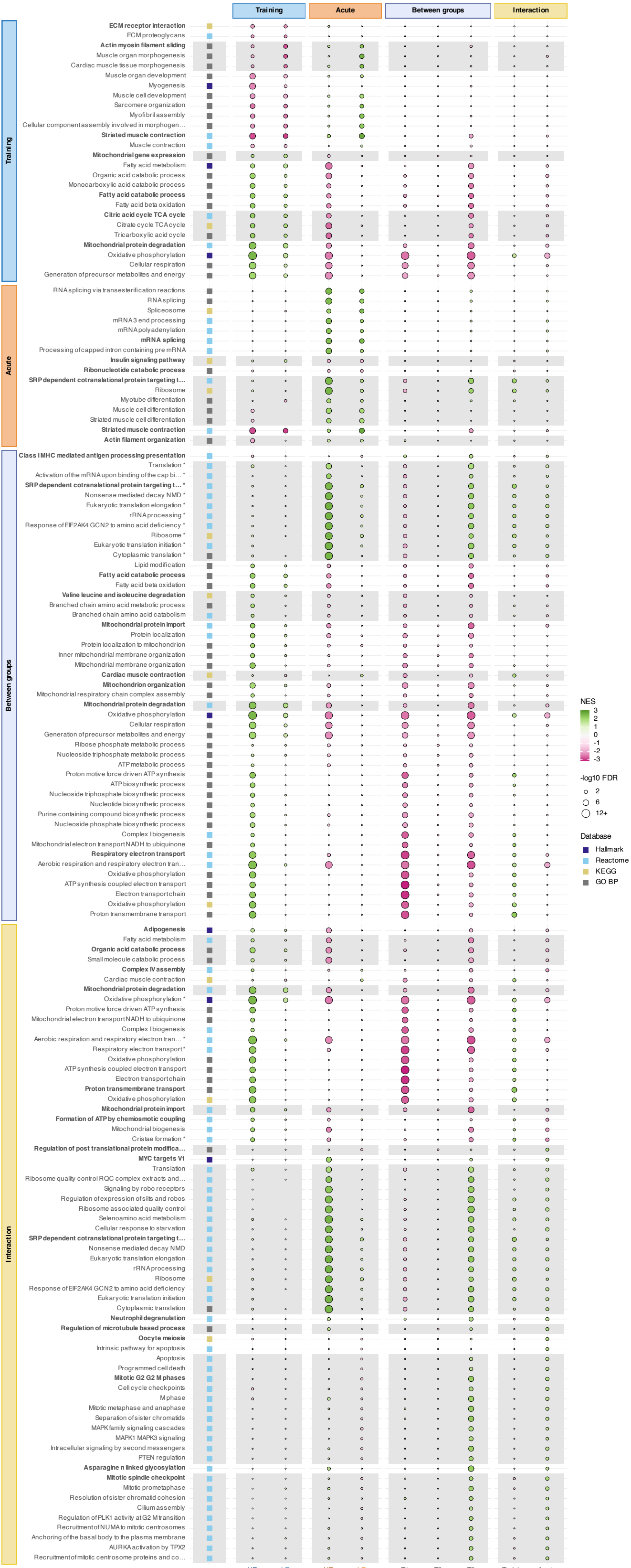
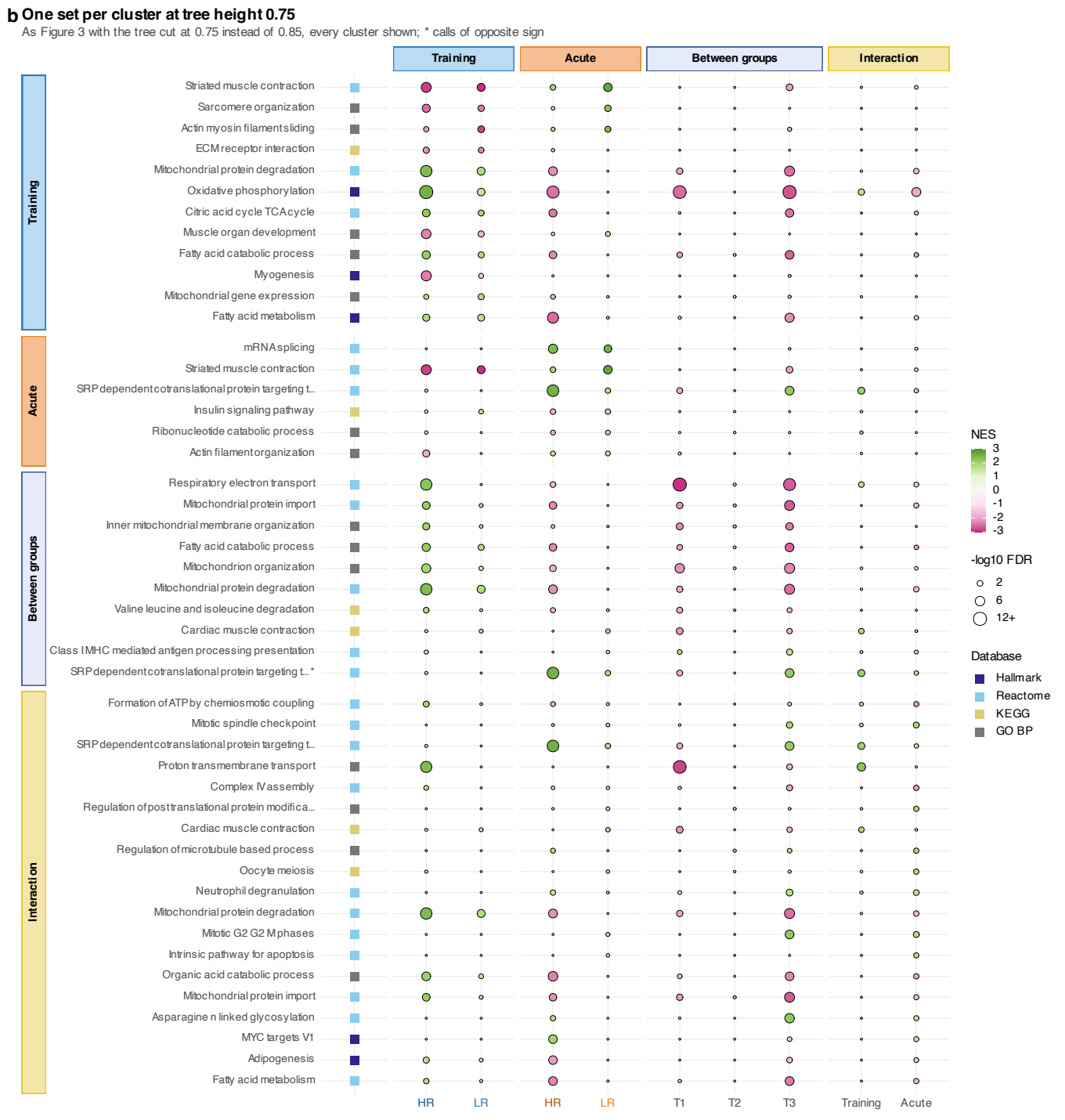
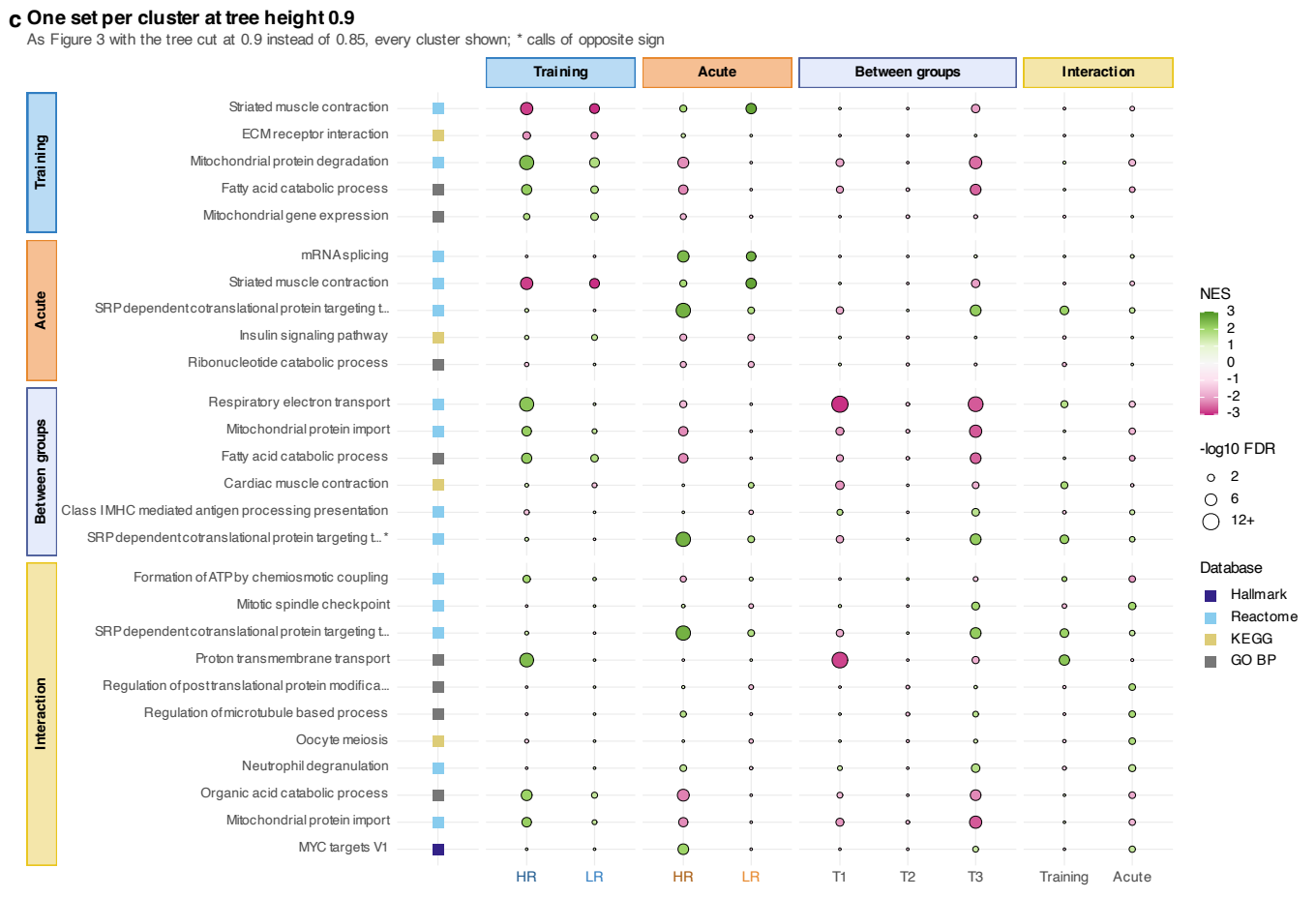
a Every candidate set, grouped by tree

Sets in leaf order of each window's tree (shared proteins, Jaccard, average linkage); bands alternate between clusters at height 0.85
bold: the cluster's representative, drawn in Figure 3; * calls of opposite sign




b One set per cluster at tree height 0.75
As Figure 3 with the tree cut at 0.75 instead of 0.85, every cluster shown; * calls of opposite sign
Training
Acute
Between groups
Interaction
Striated muscle contraction
Sarcomere organization
Actin myosin filament sliding
ECM receptor interaction
Mitochondrial protein degradation
Oxidative phosphorylation
Citric acid cycle TCA cycle
Muscle organ development
Fatty acid catabolic process
Myogenesis
Mitochondrial gene expression
Fatty acid metabolism
mRNA splicing
Striated muscle contraction
SRP dependent cotranslational protein targeting t...
Insulin signaling pathway
Ribonucleotide catabolic process
Actin filament organization
Respiratory electron transport
Mitochondrial protein import
Inner mitochondrial membrane organization
Fatty acid catabolic process
Mitochondrion organization
Mitochondrial protein degradation
Valine leucine and isoleucine degradation
Cardiac muscle contraction
Class I MHC mediated antigen processing presentation
SRP dependent cotranslational protein targeting t...*
Formation of ATP by chemiosmotic coupling
Mitotic spindle checkpoint
SRP dependent cotranslational protein targeting t...
Proton transmembrane transport
Complex IV assembly
Regulation of post translational protein modifica...
Cardiac muscle contraction
Regulation of microtubule based process
Oocyte meiosis
Neutrophil degranulation
Mitochondrial protein degradation
Mitotic G2 G2 M phases
Intrinsic pathway for apoptosis
Organic acid catabolic process
Mitochondrial protein import
Asparagine n linked glycosylation
MYC targets V1
Adipogenesis
Fatty acid metabolism
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP


c One set per cluster at tree height 0.9
As Figure 3 with the tree cut at 0.9 instead of 0.85, every cluster shown; * calls of opposite sign
Training
Acute
Between groups
Interaction
Striated muscle contraction
ECM receptor interaction
Mitochondrial protein degradation
Fatty acid catabolic process
Mitochondrial gene expression
mRNA splicing
Striated muscle contraction
SRP dependent cotranslational protein targeting t...
Insulin signaling pathway
Ribonucleotide catabolic process
Respiratory electron transport
Mitochondrial protein import
Fatty acid catabolic process
Cardiac muscle contraction
Class I MHC mediated antigen processing presentation
SRP dependent cotranslational protein targeting t...*
Formation of ATP by chemiosmotic coupling
Mitotic spindle checkpoint
SRP dependent cotranslational protein targeting t...
Proton transmembrane transport
Regulation of post translational protein modifica...
Regulation of microtubule based process
Oocyte meiosis
Neutrophil degranulation
Organic acid catabolic process
Mitochondrial protein import
MYC targets V1
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
3
2
1
0
-1
-2
-3
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP


d GO Slim themes in the nine contrasts
Every GO Slim theme; rows ordered by average-linkage clustering of their NES
Training
Acute
Between groups
Interaction
GO Slim
Detoxification
Modified amino acid metabolic process
Mitochondrion organization
Transmembrane transport
DNA replication
Membrane organization
Carbohydrate metabolic process
Sulfur compound metabolic process
Mitochondrial gene expression
Carbohydrate derivative metabolic process
Lipid metabolic process
Amino acid metabolic process
Generation of precursor metabolites and energy
Nucleobase-containing small molecule metabolic pr...
Nervous system process
Chromatin organization
Circulatory system process
Extracellular matrix organization
Mitotic cell cycle
Protein catabolic process
Lysosome organization
Cell adhesion
Cell junction organization
Cell differentiation
DNA repair
Regulation of DNA-templated transcription
Cell motility
Immune system process
Muscle system process
Cytoskeleton organization
Establishment or maintenance of cell polarity
Microtubule-based movement
Protein folding
Protein maturation
Autophagy
Regulatory ncRNA-mediated gene silencing
Nucleocytoplasmic transport
Protein-containing complex assembly
Inflammatory response
Protein localization to plasma membrane
Cytokinesis
DNA-templated transcription
Defense response to other organism
Mitotic nuclear division
Programmed cell death
Chromosome segregation
Vesicle-mediated transport
Glycoprotein biosynthetic process
Cilium organization
Intracellular protein transport
Cytoplasmic translation
tRNA metabolic process
Reproductive process
Ribosome biogenesis
mRNA metabolic process
Telomere organization
DNA recombination
Wound healing
HR
LR
HR
LR
T1
T2
T3
Training
Acute
NES
2
1
0
-1
-2
-log10 FDR
2
6
12+
Database
GO Slim